

Midterm Review Lecture STAT 286

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Outline

- 1 Introduction
- 2 Permutation Test
- 3 Average Treatment Effects
- 4 Linear Regression

Causal Inference Pipeline

- **Setup:** Binary treatment $T_i \in \{0, 1\}$; outcome Y_i may be binary, count, or continuous.

- **Goal:** Quantify the causal effect of T on Y .

- **Estimand:**

$$\tau_{\text{SATE}} = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)), \quad \tau_{\text{PATE}} = \mathbb{E}[Y(1) - Y(0)]$$

- **Identification:** A causal estimand τ is *identified* if there exists a known map g such that

$$\tau = g(\mathbb{P}), \quad \mathbb{P} \text{ is the observable law of } (Y, T, X).$$

Requires assumptions!

- **Example (under CRD):**

$$\mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y | T=1] - \mathbb{E}[Y | T=0].$$

- **Estimation & inference:** Estimate the identified estimand (permutation + test inversion, diff-in-means + Neyman variance, OLS + HC2), report s.e./CI, hope its significant

- **Causal ordering.** $T_i \rightarrow Y_i$
- **Stable Unit Treatment Value Assumption (SUTVA).**
 - *No interference:* $Y_i(T_1, \dots, T_N) = Y_i(T_i)$
 - *Consistency:* $Y_i = Y_i(T_i)$
- **Randomization.**
 - *Unconfoundedness:* $(Y_i(1), Y_i(0)) \perp\!\!\!\perp T_i$
 - *Overlap:* $0 < \Pr(T_i = 1) < 1$, in CRD: $\Pr(T_i = 1) = \frac{n_1}{n}$

When Assumptions Fail

Scenario: Effect of a flu shot (T) on days sick (Y).

- **Causal ordering:** People vaccinate *after* feeling ill $\Rightarrow Y \rightarrow T$.
- **No interference:** Herd immunity; Y_i depends on others' T_j .
- **Consistency:** “Flu shot” has different versions $\Rightarrow T=1$ not well-defined.
- **Unconfoundedness:** Health-consciousness or access affects both T and Y (selection bias).
- **Overlap:** Some strata always/never vaccinate; no comparable counterfactuals.

Result: $\mathbb{E}[Y(1) - Y(0)] \neq \mathbb{E}[Y \mid T=1] - \mathbb{E}[Y \mid T=0]$

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Permutation Test (1/4)

- **Setup:**

- Units: $i = 1, \dots, n$
- Treatment: $T_i \in \{0, 1\}$
- Outcome: $Y_i = Y_i(T_i)$, $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$
- Completely Randomized Design (CRD): $\Pr(T_i = 1 \mid \mathcal{O}_n) = \Pr(T_1 = 1) = \frac{n_1}{n}$

- **Sharp Null:** $H_0 : Y_i(1) - Y_i(0) = \tau_0$ (often $\tau_0 = 0$)

- **Test Statistic:** $S_{\tau_0} = \frac{1}{n_1} \sum_{i=1}^n T_i Y_i(T_i) - \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) (Y_i(T_i) + \tau_0)$
(many possibilities, calibrate $\mathbb{E}[S_{\tau_0} \mid H_0] = 0$ to make inference easier (not necessary))

- **Inference:** Permutation with inversion

Permutation Test (2/4)

- **Sharp Null:** $H_0 : Y_i(1) - Y_i(0) = \tau_0$
- **Test Statistic:**

$$S_{\tau_0} = \frac{1}{n_1} \sum_{i=1}^n T_i Y_i(T_i) - \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) (Y_i(T_i) + \tau_0)$$

Pipeline.

- Draw $b = 1, \dots, B$ random assignments $\mathbf{T}^{(b)} \in \{0, 1\}^n$.
- Compute potential outcomes under each assignment:
 - If $T_i^{(b)} = T_i$: $Y_i(T_i^{(b)}) = Y_i(T_i)$
 - If $(T_i^{(b)}, T_i) = (0, 1)$: $Y_i(T_i^{(b)}) = Y_i(0) = Y_i(1) - \tau_0 = Y_i(T_i) - \tau_0$
 - If $(T_i^{(b)}, T_i) = (1, 0)$: $Y_i(T_i^{(b)}) = Y_i(1) = Y_i(0) + \tau_0 = Y_i(T_i) + \tau_0$
- Recalculate the test statistic:

$$S^{(b)} = \frac{1}{n_1} \sum_{i=1}^n T_i^{(b)} Y_i(T_i^{(b)}) - \frac{1}{n_0} \sum_{i=1}^n (1 - T_i^{(b)}) (Y_i(T_i^{(b)}) + \tau_0).$$

Permutation Test (3/4)

p-values.

- $H_1 : \tau \neq \tau_0$

$$p_{\text{perm}}(\tau_0) = \frac{1 + \sum_{b=1}^B \mathbb{1}\{|S^{(b)} - \mu| \geq |S_{\tau_0}^{\text{obs}} - \mu|\}}{B + 1}$$

($\mu = \mathbb{E}[S_{\tau_0} \mid H_0(\tau_0)]$; here $\mu = 0$; alternative $p_{\text{perm}}(\tau_0) = 2 \cdot \min(p_{\text{perm},+}(\tau_0), p_{\text{perm},-}(\tau_0))$)

- $H_1 : \tau > \tau_0$

$$p_{\text{perm},+}(\tau_0) = \frac{1 + \sum_{b=1}^B \mathbb{1}\{S^{(b)} \geq S_{\tau_0}^{\text{obs}}\}}{B + 1}.$$

- $H_1 : \tau < \tau_0$

$$p_{\text{perm},-}(\tau_0) = \frac{1 + \sum_{b=1}^B \mathbb{1}\{S^{(b)} \leq S_{\tau_0}^{\text{obs}}\}}{B + 1}.$$

Confidence set.

$$\mathcal{C}_{1-\alpha} = \{\tau : p(\tau) > \alpha\},$$

where $p(\tau)$ is the p-value for $H_0(\tau)$. If the test statistic is monotone in τ , then $\mathcal{C}_{1-\alpha}$ is an interval.

Point estimate.

$$\hat{\tau} = \arg \max_{\tau} p(\tau).$$

Permutation Test (4/4)

- **Pros:** Y_i can be binary, categorical, or continuous
- **Cons:** May be less powerful; sensitive to outliers; rank-based versions mitigate this (although they require non-binary Y_i)

Rank-Sum Test (1/2)

- **Setup:** Y_i non-binary (typically continuous).
- **Sharp null:** $H_0 : Y_i(1) - Y_i(0) = \tau_0$.
- **Test statistic:** $S_{\tau_0} = \sum_{i=1}^n T_i R_i(\mathbf{Y} - \tau \mathbf{T})$.

Invariance under H_0 : Let $\mathbf{T}, \mathbf{A} \in \{0, 1\}^n$,

$$\mathbf{Y}(\mathbf{T}) - \tau_0 \mathbf{T} = \mathbf{Y}(0) = \mathbf{Y}(\mathbf{A}) - \tau_0 \mathbf{A}.$$

- $T_i = 1$: $Y_i(T_i) - \tau_0 T_i = Y_i(1) - \tau_0 = Y_i(0)$.
- $T_i = 0$: $Y_i(T_i) - \tau_0 T_i = Y_i(0)$.

Hence S_{τ_0} does not depend on τ_0 given H_0 ; use a normal approximation or permutation inference.

Permutation Inference: Draw $b = 1, \dots, B$ random assignments $\mathbf{T}^{(b)} \in \{0, 1\}^n$.

$$S^{(b)} = \sum_{i=1}^n T_i^{(b)} R_i(\mathbf{Y} - \tau \mathbf{T})$$

Note that we do not change the expression inside the ranks.

Other important facts

- *Monotonicity:* S_τ is nonincreasing in τ ; if $\tau_1 < \tau_2$ then $S_{\tau_1} > S_{\tau_2}$.
- *Maximal treatment effects:* $H_0^{\max}(\tau_0) : \max_i \{Y_i(1) - Y_i(0)\} \leq \tau_0$.

$$\Pr(S_{\tau_0} \geq s \mid H_0^{\max}(\tau_0)) \leq \Pr(S_{\tau_0} \geq s \mid H_0(\tau_0)).$$

Same one-sided sharp-null statistic yields valid (conservative) p-values.

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Finite vs. Super - Population (1/3)

	Finite-Population	Super-Population
<i>Population</i>	Set of units in the study is fixed and finite	Units are drawn i.i.d. from a large or infinite "super-population."
<i>Potential Outcomes</i>	Fixed set $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$	$(Y_i(0), Y_i(1))$ i.i.d. $\sim \mathbb{P}^*$
<i>Inference</i>	Treatment effects for this specific group of units	Treatment effects for the underlying population distribution
<i>Source of Randomness</i>	Treatment assignment	Treatment assignment Sampling of units from the population
<i>Random Variables</i>	Treatment T_i	Treatment and Potential Outcomes $T_i, Y_i(0), Y_i(1)$
<i>Example</i>	What is the ATE on these n units in our experiment?	What is the ATE in the population from which these n units were sampled?
<i>Validity</i>	Internal	External

Finite vs. Super - Population (2/3)

	Finite-Population	Super-Population
<i>Estimand</i>	$\tau_{\text{SATE}} = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))$	$\tau_{\text{PATE}} = \mathbb{E}[Y(1) - Y(0)]$
<i>Estimator</i>	Difference in means $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$	
<i>Unbiasedness</i>	$\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \tau_{\text{SATE}}$	$\mathbb{E}[\hat{\tau}] = \tau_{\text{PATE}}$
<i>Variance of Estimator</i>	$\text{Var}(\hat{\tau} \mid \mathcal{O}_n) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_*^2}{n} \leq \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0}$	$\text{Var}(\hat{\tau}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}$
<i>Estimator of Variance</i>	$\widehat{\text{Var}}(\hat{\tau}) = \frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}$ Conservative for FP, Unbiased for SP	
<i>CI in practice</i>	$\text{s.e.} = \sqrt{\frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}}$ (1 - α) $\times$ 100% CI: $[\hat{\tau} \pm z_{1-\alpha/2} \cdot \text{s.e.}]$	

Finite vs. Super - Population (3/3)

Notation

$$\begin{aligned} n_1 &= \sum_{i=1}^n T_i, \quad n_0 = n - n_1, \quad \overline{Y(t)} = \frac{1}{n} \sum_{i=1}^n Y_i(t), \quad \overline{Y}_t = \frac{1}{n_t} \sum_{i: T_i=t} Y_i, \\ S_t^2 &= \frac{1}{n-1} \sum_{i=1}^n (Y_i(t) - \overline{Y(t)})^2, \quad S_{01} = \frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - \overline{Y(1)})(Y_i(0) - \overline{Y(0)}), \\ \sigma_t^2 &= \text{Var}(Y_i(t)) = \mathbb{E}[S_t^2], \quad \hat{\sigma}_t^2 = \frac{1}{n_t-1} \sum_{i: T_i=t} (Y_i - \overline{Y}_t)^2. \end{aligned}$$

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Potential Outcomes as Linear Models

Setup. Binary treatment $T_i \in \{0, 1\}$. Define

$$\alpha_i := Y_i(0), \quad \beta_i := Y_i(1) - Y_i(0), \quad \alpha := \mathbb{E}[Y_i(0)], \quad \beta := \mathbb{E}[Y_i(1) - Y_i(0)].$$

Decomposition:

$$\begin{aligned} Y_i(t) &= Y_i(0) + (Y_i(1) - Y_i(0)) t \\ &= \mathbb{E}[Y_i(0)] + (Y_i(0) - \mathbb{E}[Y_i(0)]) \\ &\quad + \mathbb{E}[Y_i(1) - Y_i(0)] t + (Y_i(1) - Y_i(0) - \mathbb{E}[Y_i(1) - Y_i(0)]) t \\ &= \alpha + (\alpha_i - \alpha) + \beta t + (\beta_i - \beta) t \\ &= \alpha + \beta t + \underbrace{(\beta_i - \beta) t + (\alpha_i - \alpha)}_{=: \varepsilon_i(t)} \\ &= \alpha + \beta t + \varepsilon_i(t). \end{aligned}$$

Key facts. $\mathbb{E}[\varepsilon_i(t)] = 0$. Potential outcomes can be written as linear models without assumptions. Regressing $Y_i \sim [1, T_i]$ targets $\beta = \mathbb{E}[Y_i(1) - Y_i(0)]$. Extends to *categorical* T . For *continuous* T , writing $Y_i(t) = \alpha_i + \beta_i t$ imposes linearity in t .

Review: Linear Regression (1/2)

- **Linear model:**

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}^* + \boldsymbol{\varepsilon}.$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{K1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{Kn} \end{bmatrix}, \quad \boldsymbol{\beta}^* = \begin{bmatrix} \beta_0^* \\ \vdots \\ \beta_K^* \end{bmatrix}.$$

Note that since we have not specified $\boldsymbol{\varepsilon}$, this model is fully general.

- **Ordinary Least Squares (OLS) estimator:**

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2,$$

which leads to the closed-form solution:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i Y_i \right),$$

where $\mathbf{x}_i = [1, X_{1i}, \dots, X_{Ki}]^\top$. **Interpretation:** The OLS estimator gives the *best linear approximation* of $\mathbb{E}[Y_i | \mathbf{x}_i]$.

Review: Linear Regression (2/2)

Assumption: Strict exogeneity

$$\mathbb{E}[\varepsilon_i \mid \mathbf{X}_i] = 0.$$

Note: Under CRD with $\mathbf{X}_i = [1, T_i]$ this holds. If covariates are included, this does not hold necessarily, but imposes a linearity assumption (see pooled case later on).

- **Unbiasedness:**

$$\mathbb{E}[\hat{\beta} \mid \mathbf{X}] = \beta^*.$$

- **Finite-sample variance (sandwich form):**

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{\text{Bread}} \underbrace{\mathbf{X}^\top \boldsymbol{\Omega} \mathbf{X}}_{\text{Meat}} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{\text{Bread}},$$

where

$$\boldsymbol{\Omega} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2), \quad \sigma_i^2 = \text{Var}(\varepsilon_i \mid \mathbf{X}_i).$$

This is the foundation for all robust (EHW/HC2) and homoskedastic OLS variance formulas.

Homoskedasticity

- **Assumption:** Errors have constant variance

$$\text{Var}(\varepsilon_i \mid \mathbf{X}_i) = \sigma^2.$$

Then

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Omega} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

This assumption is often unrealistic in applied settings.

- **Example** (binary treatment, CRD): Let $\mathbf{X}_i = [1, T_i]^\top$. Then

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \left(\begin{bmatrix} \sum_i 1 & \sum_i T_i \\ \sum_i T_i & \sum_i T_i^2 \end{bmatrix} \right)^{-1} = \frac{1}{n_1 n_0} \begin{bmatrix} n_1 & -n_1 \\ -n_1 & n \end{bmatrix}.$$

Hence

$$\text{Var}(\hat{\beta}_T \mid \mathbf{X}) = \sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_0} \right).$$

Only valid if treatment and control variances are equal:

$$\text{Var}(Y(1)) = \text{Var}(Y(0)) = \sigma^2.$$

Heteroskedasticity: Eicker–Huber–White (EHW, HC0)

- **Motivation:** Allow error variances to differ across i .

- **HC0 (EHW) estimator:**

$$\widehat{\text{Var}}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{diag}(\hat{\varepsilon}_i^2) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1},$$

where $\hat{\varepsilon}_i = Y_i - \mathbf{X}_i^\top \hat{\beta}$.

- **Example** (binary treatment, CRD)

$$\text{Var}(\hat{\beta}_T \mid \mathbf{X}) = \frac{\tilde{\sigma}_1^2}{n_1} + \frac{\tilde{\sigma}_0^2}{n_0},$$

where

$$\tilde{\sigma}_t^2 = \frac{1}{n_t} \sum_{i: T_i=t} (Y_i - \bar{Y}_t)^2.$$

- This estimator is typically too small (biased downward). Following Neyman's correction, we would like to have

$$\hat{\sigma}_t^2 = \frac{1}{n_t - 1} \sum_{i: T_i=t} (Y_i - \bar{Y}_t)^2.$$

HC0 is *consistent* but biased in finite samples.

Heteroskedasticity: HC2

- **Idea:** Correct HC0's downward bias using leverage adjustment.
- **HC2 estimator:**

$$\widehat{\text{Var}}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{diag}\left(\frac{\hat{\varepsilon}_i^2}{1 - p_{ii}}\right) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1},$$

where $p_{ii} = \mathbf{X}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_i$ is the leverage of observation i .

- **In a CRD with $\mathbf{X}_i = [1, T_i]^\top$:**

$$p_{ii} = \frac{1}{n_1 n_0} \begin{bmatrix} 1 \\ T_i \end{bmatrix}^\top \begin{bmatrix} n_1 & -n_1 \\ -n_1 & n \end{bmatrix} \begin{bmatrix} 1 \\ T_i \end{bmatrix} = \frac{1}{n_1} T_i + \frac{1}{n_0} (1 - T_i).$$

Thus

$$\frac{1}{1 - p_{ii}} = \begin{cases} \frac{n_1}{n_1 - 1}, & T_i = 1, \\ \frac{n_0}{n_0 - 1}, & T_i = 0. \end{cases}$$

This correction yields the **Neyman variance**:

$$\text{Var}(\hat{\beta}_T \mid \mathbf{X}) = \frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}, \quad \hat{\sigma}_t^2 = \frac{1}{n_t - 1} \sum_{i: T_i = t} (Y_i - \bar{Y}_t)^2.$$

In very small samples, adjust further by scaling with $\frac{n}{n - (K+1)}$ or use **HC3**.

Linear Regression with No Covariates

Setup. Recall $Y_i(t) = \alpha + \beta t + \varepsilon_i(t)$. Regressing $Y \sim [1, T]$ yields (without assumptions)

$$\hat{\alpha} = \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i = \bar{Y}_0,$$

$$\hat{\beta}_{\text{DiM}} = \frac{1}{n_1} \sum_{i=1}^n T_i Y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i.$$

Interpretation. $\hat{\beta}_{\text{DiM}}$ is the difference-in-means (DiM) estimator.

Variance. Using HC2 variance estimation (Neyman's formula),

$$\widehat{\text{Var}}(\hat{\beta}_{\text{DiM}}) = \frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}, \quad \hat{\sigma}_t^2 = \frac{1}{n_t - 1} \sum_{i: T_i=t} (Y_i - \bar{Y}_t)^2.$$

Thus, OLS with an intercept and binary T_i reproduces the classical difference-in-means (DiM) estimator.

Linear Regression with Covariates (Pooled) (1/3)

Motivation. Even under randomization, covariates can be imbalanced; adjusting for X can improve precision.

Pooled (ANCOVA) model.

$$Y_i = \alpha + \beta T_i + \gamma^\top \mathbf{X}_i + \varepsilon_i.$$

Interpretation (parallel slopes).

$$T_i = 1: \quad Y(1) = \alpha + \beta + \gamma^\top \mathbf{X}_i + \varepsilon_i,$$

$$T_i = 0: \quad Y(0) = \alpha + \gamma^\top \mathbf{X}_i + \varepsilon_i.$$

Key assumption: a common slope γ for both groups.

Linear Regression with Covariates (Pooled) (2/3)

Normal equations (univariate X). OLS residuals are orthogonal to each regressor:

$$\sum_{i=1}^n \hat{\varepsilon}_i = 0, \quad \sum_{i=1}^n T_i \hat{\varepsilon}_i = 0, \quad \sum_{i=1}^n X_i \hat{\varepsilon}_i = 0.$$

From the first two,

$$\bar{Y} = \hat{\alpha} + \hat{\beta} \frac{n_1}{n} + \hat{\gamma} \bar{X}, \quad \bar{Y}_1 = \hat{\alpha} + \hat{\beta} + \hat{\gamma} \bar{X}_1.$$

(Equivalently, using $\sum (1 - T_i) \hat{\varepsilon}_i = 0$: $\bar{Y}_0 = \hat{\alpha} + \hat{\gamma} \bar{X}_0$.)

Eliminate $\hat{\alpha}$:

$$\hat{\beta} = (\bar{Y}_1 - \bar{Y}_0) - \hat{\gamma} (\bar{X}_1 - \bar{X}_0) = \hat{\beta}_{\text{DiM}} - \hat{\gamma} (\bar{X}_1 - \bar{X}_0).$$

Bias (finite sample). With pre-treatment X and CRD,

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}[Y(1) - Y(0)] - \underbrace{\mathbb{E}[\hat{\gamma} (\bar{X}_1 - \bar{X}_0)]}_{\neq 0, \text{Cov}(\hat{\gamma}, \bar{X}_1 - \bar{X}_0) \neq 0} \neq \mathbb{E}[Y(1) - Y(0)].$$

Under random sampling and CRD, $\bar{X}_1 - \bar{X}_0 = O_p(n^{-1/2})$ so the bias vanishes as $n \rightarrow \infty$.

Linear Regression with Covariates (Pooled) (3/3)

Variance (heuristic). If $\mathbb{E}[Y \mid T, X] = \alpha + \beta T + \gamma X$ and homoskedasticity holds,

$$\text{Var}(\hat{\beta}_P \mid T, X) \approx \text{Var}(\hat{\beta}_{\text{DiM}} \mid T, X) (1 - R_{Y \sim X}^2),$$

where $R_{Y \sim X}^2$ is from regressing Y on $(1, X)$. Thus variance decreases with predictive power of X ; without correct specification this reduction is not guaranteed.

Takeaway.

- $\hat{\beta}_P \rightarrow \beta$ (consistent), but not necessarily unbiased in finite samples.
- If $\mathbf{X}$ is predictive of Y , this regression can reduce variance.
- However, if the model is misspecified (e.g., treatment and control have different covariate effects), efficiency gains may be lost or even reversed.

Linear Regression with Covariates (Interacted) (1/2)

Idea. The pooled model assumes equal slopes for treatment and control. To relax this, we allow interaction terms between treatment and covariates.

$$Y_i = \alpha + \beta T_i + \gamma^\top \tilde{\mathbf{X}}_i + \delta^\top (T_i \tilde{\mathbf{X}}_i) + \varepsilon_i,$$

where $\tilde{\mathbf{X}}_i = \mathbf{X}_i - \overline{\mathbf{X}}$ (demeaned covariates).

Interpretation.

$$T_i = 1: Y(1) = \alpha + \beta + \delta^\top \tilde{\mathbf{X}}_i + \varepsilon_i,$$

$$T_i = 0: Y(0) = \alpha + \gamma^\top \tilde{\mathbf{X}}_i + \varepsilon_i.$$

Treatment and control now have different intercepts and covariate effects.

Imputation form. The estimated treatment effect can be written as

$$\hat{\beta}_1 = \frac{1}{n} \sum_{i=1}^n [T_i(Y_i - \hat{Y}_i(0)) + (\hat{Y}_i(1) - Y_i)(1 - T_i)],$$

where

$$\hat{Y}_i(t) = \hat{\alpha} + \hat{\beta}_1 t + \hat{\gamma}^\top \tilde{\mathbf{X}}_i + \hat{\delta}^\top (t \tilde{\mathbf{X}}_i).$$

Linear Regression with Covariates (Interacted) (2/2)

Assumptions: SUTVA, complete randomization (CRD), and random sampling.

Main results:

- **Consistency:**

$$\hat{\beta}_1 \xrightarrow{P} \tau = \mathbb{E}[Y_i(1) - Y_i(0)].$$

- **Asymptotic normality:**

$$\sqrt{n}(\hat{\beta}_1 - \tau) \xrightarrow{d} \mathcal{N}(0, \sigma_1^2).$$

- **Efficiency:**

$$\sigma_1^2 \leq \sigma_{\text{DiM}}^2 \quad \text{and} \quad \sigma_1^2 \leq \sigma_{\text{P}}^2.$$

That is, the interacted regression is at least as efficient as DiM or pooled OLS.

- **Variance estimation:** The HC2 (heteroskedasticity-robust) variance estimator is consistent for $\hat{\beta}_1$.

Key insight: These results hold *even if the regression model is misspecified*, the interacted (Lin, 2013) approach remains valid under randomization.

Any Questions?

Good luck!

You've got this! Go show off those causal superpowers.